



An experimental study on fairness-aware machine learning for credit scoring problems

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Received: 15 December 2025 / Accepted: 28 May 2026
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Abstract

The digitalization of credit scoring has become essential for financial institutions and commercial banks, especially in the era of digital transformation. Machine learning (ML) techniques are commonly used to evaluate customers' creditworthiness. However, the predicted outcomes of ML models can be biased toward protected attributes, such as race or gender. Numerous fairness-aware ML models and fairness measures have been proposed in recent years. However, their behavior in the context of credit scoring has not been thoroughly investigated. In this paper, we present a comprehensive experimental study of fairness-aware ML for credit scoring. Our study examines several key aspects of the problem, including financial datasets, predictive models, and fairness measures. In addition, we analyze structural dependencies between protected attributes and the class label using a Bayesian network to better understand statistical relationships within the datasets. We further provide a detailed evaluation of fairness-aware predictive models and fairness measures on widely used credit scoring datasets. The experimental results show that fairness-aware models achieve a better balance between predictive accuracy and fairness than traditional classification models.

Keywords Credit scoring · Fairness · Fairness measures · Financial dataset · Machine learning · Predictive model

JEL Classification D63 · G21 · C53 · C45

1 Introduction

Traditional banking systems are time-consuming, require substantial human resources, and involve complex manual procedures. There is a risk that traditional banks will become obsolete due to technological advancements. Therefore, a digi-

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tal transformation in the banking systems is needed to fasten and ease banking tasks. We have seen the new technologies in the banking sector as vivid examples of digital transformation, such as Robotic Process Automation (RPA), Big Data, Cloud Computing, and Blockchain.¹ In the digital transformation process, the automation of banking procedures is an essential requirement of banks. Hence, the digitalization of credit scoring is also essential because credit scoring is a crucial phase in the risk management process of financial organizations and commercial banks. In order to automatically perform customer credit scoring, a variety of ML methods have been applied effectively (Bhatore et al., 2020; Dastile et al., 2020; Dumitrescu et al., 2022; Trivedi, 2020). The experimental results are calculated based on existing customers' financial and non-financial data at the time of credit scoring and customer rating.

However, apart from the advantages of ML techniques on credit scoring, there is growing evidence regarding the discriminatory impact of ML-based decision-making on individuals and groups based on protected attributes such as race or gender (Le Quy et al., 2022; Ntoutsu et al., 2020). Therefore, ensuring fairness with respect to protected attributes in ML models is an important requirement (Mehrabi et al., 2021; Pessach & Shmueli, 2022). It is crucial for ML models to be highly accurate while minimizing discrimination against individuals or groups of people with regard to protected attributes.

In the ML research community, fairness-aware ML has been investigated in many domains, such as finance, healthcare, and education (Le Quy et al., 2022). However, there are only a few studies on fairness-aware ML in the banking sector, particularly on the credit scoring problem. The pioneering work of Bono et al. (2021) was the first empirical study of the accuracy and statistical fairness of different credit scoring technologies in the UK context. The experiments were conducted on only a dataset collected in the UK. Then, Kozodoi et al. (2022) provided an evaluation of different fairness-aware classifiers on credit scoring datasets. However, they reported experimental results on only three fairness measures. The review of Adegoke et al. (2024) offered a thorough analysis on the fairness of credit scoring models in relation to mortgage accessibility for underserved populations. The work of Mariscal et al. (2024) focused on analyzing the trade-off between performance and fairness of several fairness-aware ML techniques. However, they reported the results on only two fairness measures (demographic parity and equalized odds). Recently, Hurlin et al. (2024) presented a framework aimed at formally testing the null hypothesis of fairness and helping lenders and regulatory bodies to identify the factors driving unfair outcomes. Nevertheless, the experimental results were reported on only the German credit dataset.

Furthermore, the choice of the fairness criterion leads to severe consequences for the social impact of lending decisions with credit scoring (Liu et al., 2018). Without constraints, a credit scoring model takes advantage of all available information, including protected attributes, and may discriminate against protected groups if doing so improves predictive performance. The goal of incorporating fairness is to modify decision-making (i.e., scoring) practices to achieve equitable, non-discriminatory outcomes. Indeed, more than 20 fairness measures have been introduced in the domain of fairness-aware ML (Verma, 2018). Therefore, selecting an appropriate fairness measure for credit scoring is not straightforward, as no single measure is universally

¹ <https://boostylabs.com/blog/digital-transformation-in-banking>

applicable (Mehrabani et al., 2021; Verma, 2018). As a result, it remains unclear how different categories of fairness-aware models behave across diverse credit datasets and fairness measures. Moreover, most prior studies focus primarily on model-level fairness evaluation and do not examine structural dependencies between protected attributes and the class label in the datasets prior to model training. Therefore, we analyze structural dependencies between protected attributes and the class label using a Bayesian network (BN), which provides a probabilistic framework for modeling relationships among variables in the data. These dependencies should be interpreted as statistical associations rather than evidence of causal discrimination.

To address the aforementioned gap, in this work, we investigate fairness-aware ML in the context of credit scoring by analyzing both statistical dependencies in credit datasets and the behavior of fairness-aware models. Our main contributions are summarized as follows:

- We review key concepts in fairness-aware ML and discuss commonly used fairness measures relevant to the credit scoring problem.
- We analyze structural dependencies between protected attributes and the class label using the BN, providing insights into statistical relationships present in commonly used credit scoring datasets.
- We conduct a systematic empirical evaluation of traditional and fairness-aware classification models across multiple public credit scoring datasets and seven widely used fairness measures.

This paper is structured as follows. Section 2 introduces fairness-aware ML techniques and models relevant to the credit scoring problem. Section 3 presents commonly used fairness measures. Section 4 describes the credit scoring datasets and analyzes potential statistical dependencies using the BN. Section 5 reports the experimental evaluation of traditional and fairness-aware models with respect to predictive performance and fairness measures. Finally, Sect. 6 concludes the paper and outlines directions for future research.

2 Fairness-aware machine learning (for credit scoring)

In this section, we provide an overview of fairness-aware predictive models designed for classification tasks in the financial domain, with a particular focus on models potentially applicable to the credit scoring problem. Three main categories of fairness-aware predictive models are reviewed, including pre-processing, in-processing, and post-processing approaches.

2.1 Formulation of the credit scoring problem

The credit scoring problem is described as “methods used for classifying applicants for credit into ‘good’ and ‘bad’ risk classes” (Dastile et al., 2020; Hand & Henley, 1997). Similarly, it can be defined as a “set of decision models and their underlying techniques that aid credit lenders in the granting of credit” (Thomas et al., 2002) or

a tool used to quantify credit risk using applicants' financial behavior and repayment history (Ayari et al., 2025). Therefore, we consider the credit scoring problem as a binary classification problem.

We denote D as a dataset with class attribute $Y = \{+, -\}$; e.g., $Y = \{\text{good credit}, \text{bad credit}\}$. A binary protected attribute is denoted by S , $S \in \{s, \bar{s}\}$ where s is the protected group and $\bar{s}$ is the non-protected group; e.g., $S = \text{"Sex"}$ and $S \in \{\text{female}, \text{male}\}$. $\hat{Y} = \{+, -\}$ is the predicted class. Hence, the protected and non-protected groups with respect to positive (negative, respectively) classes are s_+ (s_-), $\bar{s}_+$ ($\bar{s}_-$). We refer to the positive class as the target class, e.g., *good credit*.

The goal of the fairness-aware classification model in the credit scoring problem is to find a map function $f : D \mapsto Y$ that minimizes the loss and mitigates simultaneously discriminatory outcomes.

2.2 Fairness-aware ML models

Three main approaches are commonly used to mitigate bias in ML models and achieve fairness: (i) pre-processing, (ii) in-processing, and (iii) post-processing methods. (Mehrabi et al., 2021; Ntoutsis et al., 2020; Pessach & Shmueli, 2022).

In the pre-processing approach, researchers focus on the data, which are the primary source of bias. The goal is to generate a "balanced" dataset and then apply any ML algorithms to that. For example, class labels are altered, different weights are assigned to instances, or the protected and unprotected groups are balanced in the training set. Techniques such as learning fair representations (LFR) aim to encode data effectively while obscuring protected attributes (Zemel et al., 2013). Similarly, the disparate impact remover (DIR) adjusts feature values to enhance group fairness while preserving rank-ordering within groups (Feldman et al., 2015).

In-processing approaches reformulate the classification problem by explicitly incorporating the model's discrimination behavior in the objective function through regularization or constraints or by training on latent target labels. For example, according to Agarwal's method, a fair classification can be reduced to a series of cost-sensitive classification problems with the lowest (empirical) error under the desired constraints (Agarwal et al., 2018). AdaFair (Iosifidis & Ntoutsis, 2019), a sequential fair ensemble, extends AdaBoost's weighted distribution approach by taking into account the cumulative fairness of the learner up until the current boosting round and moreover, accounts for class imbalance by optimizing for balancing error instead of an overall error.

Unlike the above two approaches, post-processing methods modify the outputs of classification models after training the models. It involves altering the model's internals (white-box approaches) or its predictions (black-box approaches). White-box post-processing methods adjust the internal decision-making criteria of a model (Kamiran & Calders, 2012). For example, decision thresholds might be altered to balance outcomes across sensitive groups. This requires direct access to the model's decision rules, making it most suitable for scenarios where model transparency is available. Black-box post-processing methods, by contrast, operate solely on the

model's outputs (Kim et al., 2019). This makes them model-agnostic and widely applicable. For instance, calibrated equalized odds post-processing (CEP) optimizes calibrated classifier score outputs to determine the probabilities of altering output labels to achieve equalized odds (Pleiss et al., 2017). Similarly, equalized odds post-processing (EOP) uses linear programming to find probabilities for modifying output labels, ensuring equalized odds objectives are met (Hardt et al., 2016).

In this work, we demonstrate the performance of the three above approaches with 6 well-known models: i) Pre-processing approach: Learning fair representations (LFR), Disparate impact remover (DIR); ii) In-processing approach: Agarwal's, AdaFair; iii) Post-processing approach: Equalized odds post-processing (EOP), Calibrated equalized odds post-processing (CEP).

3 Fairness measures

We perform the evaluation on the most popular group fairness notions which are used to determine how fair the model's results are. Because no single fairness definition is universally appropriate for all applications, multiple measures are considered to provide a more comprehensive view of fairness behavior. Importantly, the fairness notions may be turned into measures by taking a difference or a ratio of the equation components (liobait, 2017). Therefore, in this paper, we use the terms "fairness notion" and "fairness measure" interchangeably. The fairness measures are chosen based on the number of citations.² In all fairness measures, a higher value indicates a larger difference in predictions between the two groups, so the model is less fair, i.e., 0 stands for no discrimination. Table 1 provides an overview of fairness measures. Fairness measures are defined as below using notations described in Sect. 2.1.

Statistical parity (SP)

$$SP = P(\hat{Y} = + \mid S = \bar{s}) - P(\hat{Y} = + \mid S = s) \quad (1)$$

Equal opportunity (EO)

$$EO = |P(\hat{Y} = - \mid Y = +, S = \bar{s}) - P(\hat{Y} = - \mid Y = +, S = s)| \quad (2)$$

Table 1 An overview of fairness measures

Fairness measures	#Citations	Values
Statistical parity (SP) (Dwork et al., 2012)	4398	$[-1, 1]$
Equal opportunity (EO) (Hardt et al., 2016)	4935	$[0, 1]$
Equalized odds (EOd) (Hardt et al., 2016)	4935	$[0, 2]$
Predictive parity (PP) (Chouldechova, 2017)	2461	$[0, 1]$
Predictive equality (PE) (Corbett-Davies et al., 2017)	1545	$[0, 1]$
Treatment equality (TE) (Berk et al., 2021)	1170	$(-\infty, \infty)$
ABROCA (Gardner et al., 2019)	191	$[0, 1]$

² Reported by Google Scholar on 21st October 2024.

Equalized odds (EOd)

$$EOd = \sum_{y \in \{+, -\}} |P(\hat{Y} = + | S = s, Y = y) - P(\hat{Y} = + | S = \bar{s}, Y = y)| \quad (3)$$

Predictive parity (PP)

$$PP = |P(Y = + | \hat{Y} = +, S = s) - P(Y = + | \hat{Y} = +, S = \bar{s})| \quad (4)$$

Predictive equality (PE)

$$PE = |P(\hat{Y} = + | Y = -, S = s) - P(\hat{Y} = + | Y = -, S = \bar{s})| \quad (5)$$

Treatment equality (TE) The treatment equality (TE) is the difference in the ratio of False Negative (FN) to False Positive (FP) between protected group (prot.) and non-protected (non-prot.) groups. TE measure becomes unbounded if $FP_{prot.}$ or $FP_{non-prot.}$ is zero.³

$$TE = \frac{FN_{prot.}}{FP_{prot.}} - \frac{FN_{non-prot.}}{FP_{non-prot.}} \quad (6)$$

*Absolute Between-ROC Area (ABROCA)*⁴ It measures the divergence between the protected (ROC_s) and non-protected group ($ROC_{\bar{s}}$) curves across all possible thresholds $t \in [0, 1]$ of false positive rates (FPR) and true positive rates (TPR). The absolute difference between the two curves is calculated to account for cases where the curves intersect.

$$\int_0^1 |ROC_s(t) - ROC_{\bar{s}}(t)| dt \quad (7)$$

4 Datasets for credit scoring

This section describes the credit scoring datasets used in this study. We perform a preliminary data analysis using the BN to identify statistical dependencies between protected attributes and the class label.

To identify the relevant datasets, we use several research databases such as Google Scholar,⁵ Paper With Code,⁶ ResearchGate,⁷ ScienceDirect⁸ with “datasets for credit

³ <https://docs.aws.amazon.com/sagemaker/latest/dg/clarify-post-training-bias-metric-te.html>

⁴ ROC: Receiver operating characteristic.

⁵ <https://scholar.google.com/>

⁶ <https://paperswithcode.com/>

⁷ <https://www.researchgate.net/>

⁸ <https://www.sciencedirect.com/>

scoring” as the primary query term to narrow down the search. We reviewed studies published between 2010 and 2021, focusing on literature on ML-based credit scoring and the increasing research interest in fairness-aware ML. The increasing use of ML in credit scoring has raised concerns about fairness, transparency, and potential discriminatory outcomes in lending decisions (Kozodoi et al., 2022; Mehrabi et al., 2021). Figure 1 illustrates the use of existing credit scoring datasets in scientific works.

We select datasets for our experiments based on several criteria: (i) The dataset must contain the protected attributes, such as gender, race, etc.; (ii) The dataset must have the “target/class” attribute which is used for classification operation; (iii) The dataset must have more than 500 instances. To this end, we employ 5 datasets for our evaluation,⁹ described in Table 2.

In the next step, inspired by the work of Le Quy et al. (2022), we perform a preliminary data analysis using the BN (Holmes, 2008). If the generated BN reveals direct or indirect connections between a protected attribute and the class label, this indicates a statistical dependency between these variables. Such dependencies may be important to consider when analyzing fairness in predictive models (Gonzalez-Sendino et al., 2024; Lam, 2024; Van Breugel et al., 2021). We also transform the numerical attributes into categorical attributes in order to reduce the computational complexity of the BN generator (Chen et al., 2017). Regarding the BN of the *Credit card clients* and *German credit* datasets, we refer to the work of Le Quy et al. (2022).

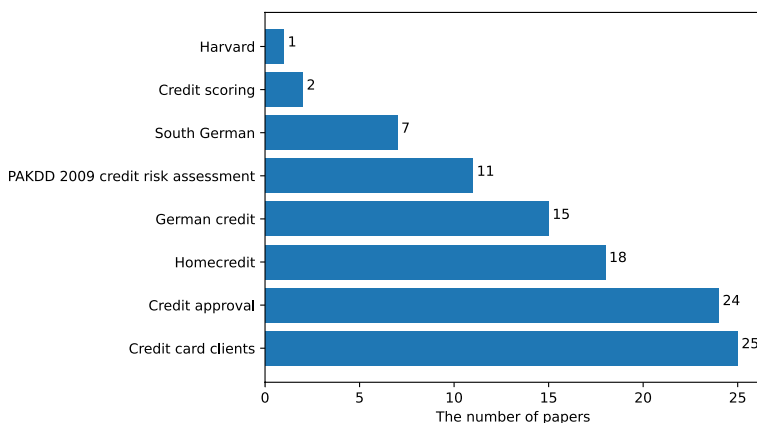


Fig. 1 The use of credit scoring datasets

⁹We use the term “Sex” to refer to “Gender”, “Marital status” to refer to “Marriage”, “Family status”. Abbreviation: IR: Imbalance ratio.

Table 2 An overview of credit scoring datasets

Datasets	#Instances	#Instances (cleaned)	#Attributes	Protected attribute(s)	Class label (positive)	IR (+:-)
Credit approval	690	678	15	Sex, Age	Approved	1:1.23
Credit card clients	30,000	30,000	23	Sex, Education, Marital status	Default payment	1:3.52
Credit scoring	8,755	8,755	17	Age, Sex, Marital status	Good credit	11.58:1
German credit	1,000	1,000	21	Age, Sex	Good credit	2.33:1
PAKDD credit	50,000	38,896	47	Age, Sex, Marital status	Bad credit	2.83:1

4.1 Credit approval dataset

The credit approval dataset¹⁰ (another name: Australian credit approval dataset) contains information of 690 credit card applications. The classification task is to predict whether an application is approved or not (class attribute: *Approved*). The positive class is approved (value +). To generate the BN, we discretize *age* = $\{< 25, \geq 25\}$; the continuous variables *Debt*, *YearsEmployed*, *CreditScore* and *Income* are encoded based on their median value: *Debt* = $\{\leq 2.875, > 2.875\}$, *YearsEmployed* = $\{\leq 1, > 1\}$, *Income* = $\{\leq 5, > 5\}$. Figure 2 depicts the BN of the credit approval dataset where the class label is highlighted in yellow, while the protected attributes are colored in blue. In the BN, there is a strong relationship between *Class* and *Bank account* attributes. The analysis shows that 79.55% of people with bank accounts (*Bank account* = “Yes”) are approved for credit, while the rate among people without bank accounts is only 5.86% (Fig. 3).

4.2 Credit card clients dataset

The credit card clients dataset¹¹ consists of information about 30,000 customers in Taiwan in October 2005 (Yeh & Lien, 2009). The prediction task is to forecast if a customer will face the default situation in the next month or not (class attribute: *Y*). The positive class is default payment (value 0).

4.3 Credit scoring dataset

The credit scoring dataset¹² has 8,755 records of customers collected by a FinTech company in Central Asia. The dataset was published on Kaggle in 2021 by Davronov. Predicting whether a customer has good credit (value 1) is the main goal (class attribute: *label*). We categorize two numerical attributes based on their median value: *INPS_mln_sum* = $\{\leq 1.7, > 1.7\}$ and *Score_point* = $\{\leq 0, > 0\}$. Figure 4 demonstrates the BN of the Credit scoring dataset. There is an indirect relationship

¹⁰ <http://archive.ics.uci.edu/ml/datasets/credit+approval>

¹¹ <https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>

¹² <https://www.kaggle.com/code/islombekdavronov/credit-scoring>

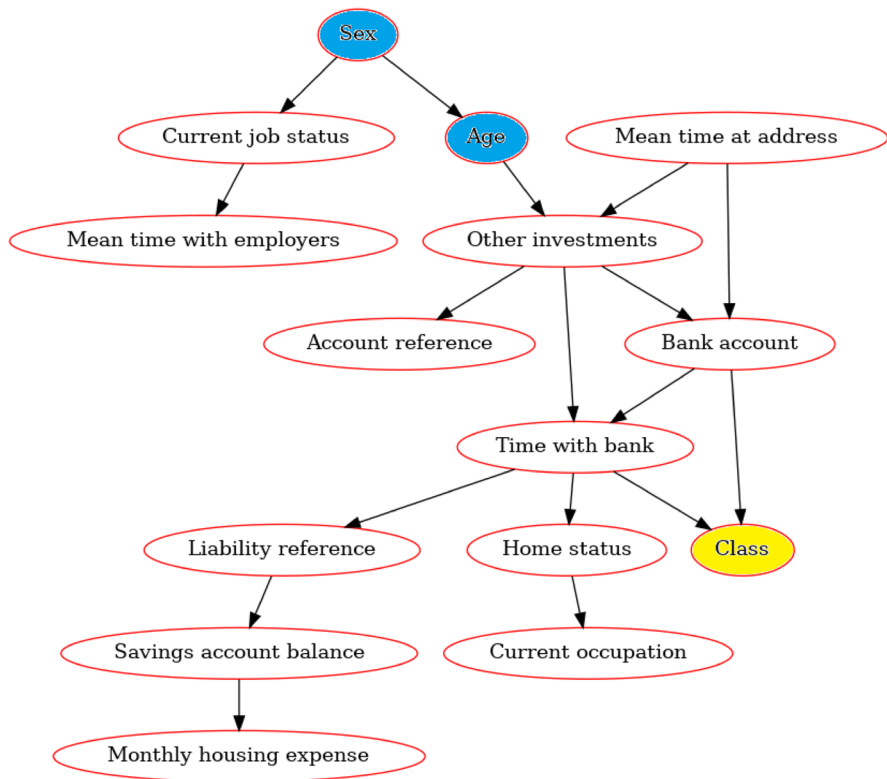


Fig. 2 Credit approval: Bayesian network (class label: *Class*, protected attributes: *Age*, *Sex*)

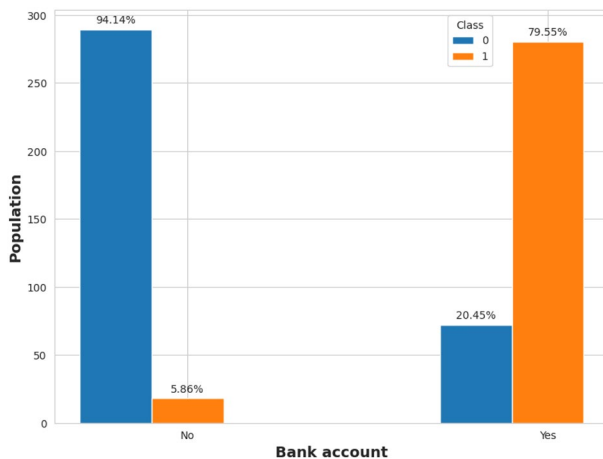


Fig. 3 Credit approval: Relationship between class label and *Bank account* attribute

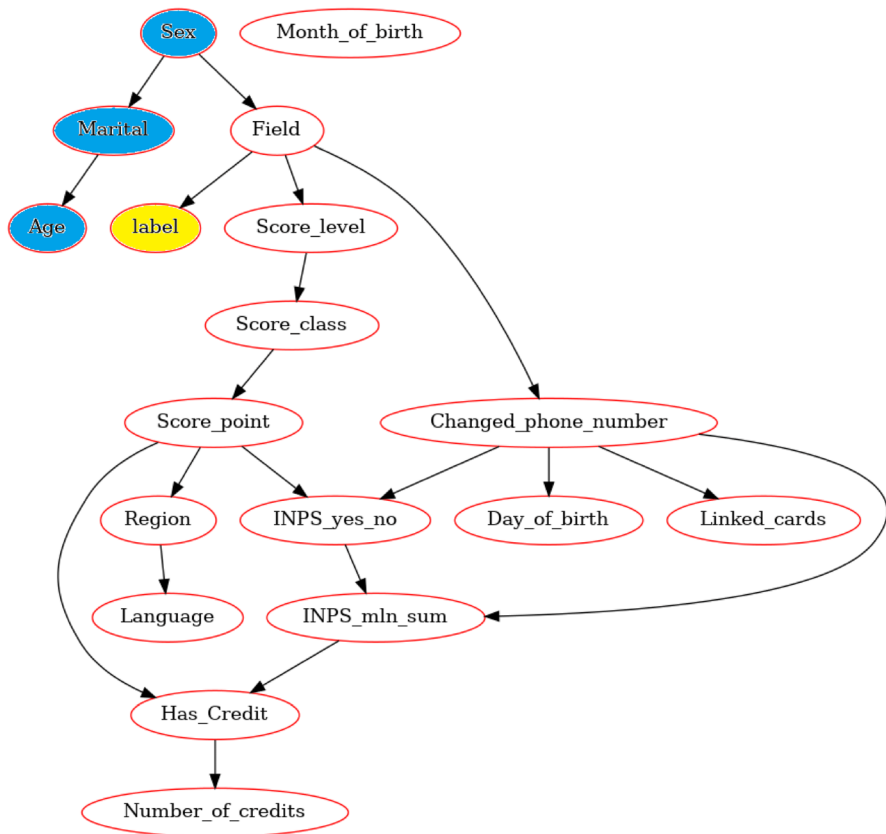


Fig. 4 Credit scoring: Bayesian network (class label: *Label*, protected attributes: *Age*, *Marital*, *Sex*)

between the *class label* and the *Sex* attribute, which indicates a statistical dependency between these variables.

4.4 German credit dataset

The German credit dataset¹³ contains observations for 1000 applicants for credit. It was published on the UCI repository website¹⁴ by Hofmann (1994). The goal is to predict whether a customer has good (value 1) or bad credit (value 2) (class attribute: *class-label*). The positive class is good (value 1).

¹³ <https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data>

¹⁴ <https://archive.ics.uci.edu/>

4.5 PAKDD 2009 credit risk assessment dataset

The PAKDD credit risk assessment dataset¹⁵ was provided by the PAKDD data mining competition in 2009 with 50,000 instances. The class label is *TARGET_LABEL_BAD*, intending to predict if a customer has bad credit. Therefore, the positive class in this dataset is set based on the value *TARGET_LABEL_BAD* = 1 (bad credit). To generate the BN, we remove attributes with too many distinct values, such as *ID_CLIENT*, *PROFESSION_CODE*, *RESIDENCIAL_PHONE_AREA_CODE*.¹⁶ Moreover, we discretize continuous attributes based on their median: *Quant_dependants* = {≤ 0, > 0}; *Months_in_residence* = {≤ 5, > 5}; *Personal_monthly_income* = {≤ 500, > 500}. Figure 5 depicts the BN of the PAKDD credit dataset. *Sex* attribute has an indirect connection with the class label. We observe that up to 53.16% of female customers have bad credit ratings when their monthly income is low (below 500). Nevertheless, even among female customers with good credit ratings, the proportion in the higher income group (above 500) remains low (Fig. 6).

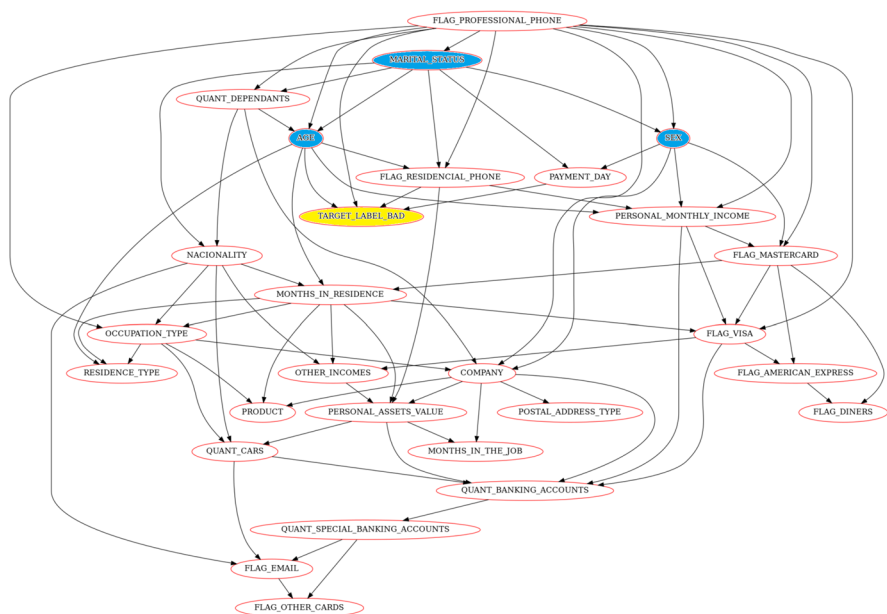


Fig. 5 PAKDD credit: Bayesian network (class label: *TARGET_LABEL_BAD*, protected attributes: *AGE*, *SEX*, *MARITAL_STATUS*)

¹⁵ <https://github.com/JLZml/Credit-Scoring-Data-Sets>

¹⁶ The names of these variables are taken from the variable description file provided with the dataset.

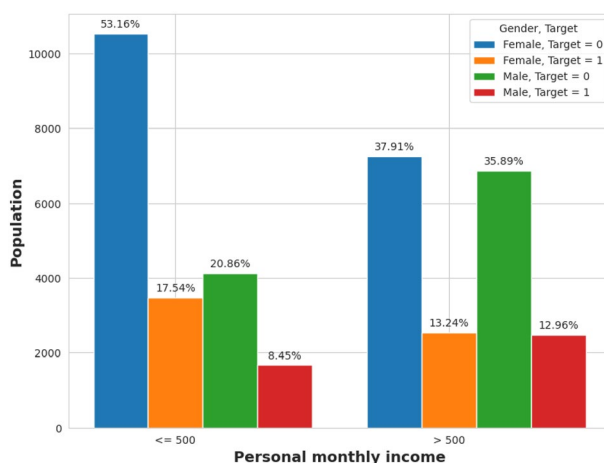


Fig. 6 PAKDD credit: Relationship between *Sex*, *Personal_monthly_income* and class label

5 Evaluation

We experiment with selected fairness-aware predictive models using prevalent fairness measures and various financial datasets to evaluate the performance of a wide range of predictive models (traditional models, pre-processing, in-processing, and post-processing fairness-aware predictive models). The preliminary results are the primary means to select the appropriate fairness-aware predictive models and fairness measures for the credit scoring problem.

5.1 Experimental setups

5.1.1 Predictive models

Traditional predictive models. We perform experiments on four well-known traditional classification models applied for the credit scoring problem (Brown & Mues, 2012; Trivedi, 2020), namely Decision Tree (DT); Naive Bayes (NB); Multi-layer Perceptron (MLP); and k-nearest neighbors (kNN).

Fairness-aware ML models. Three groups of fairness-aware ML models are chosen: i) Pre-processing approach: Learning fair representations (LFR) (Zemel et al., 2013), Disparate impact remover (DIR) (Feldman et al., 2015); ii) In-processing approach: Agarwal's (Agarwal et al., 2018), Adafair (Iosifidis & Ntoutsis, 2019); iii) Post-processing approach: Calibrated equalized odds post-processing (CEP) (Pleiss et al., 2017), Equalized odds post-processing (EOP) (Hardt et al., 2016).

5.1.2 Training and test sets

We use 70% of the data for training and 30% for testing with a single split. All traditional predictive models are implemented and executed with the default param-

Table 3 Credit approval: performance of predictive models. Protected attribute: Sex

Model	BA	Acc.	SP	EO	EOd	PP	PE	TE	ABROCA
DT	0.7646	0.7696	0.0840	0.0962	0.1167	0.0536	0.0205	0.2190	0.0378
NB	0.7629	0.7794	0.0530	0.0637	0.0729	0.0647	0.0092	-0.5357	0.0483
MLP	0.7038	0.7107	0.0922	0.0950	0.1432	0.0414	0.0482	0.2697	0.1005
kNN	0.6494	0.6617	-0.1051	<u>0.0084</u>	0.0638	0.1131	0.0554	-0.6150	0.0664
DIR-DT	0.5421	0.5196	0.0137	0.1117	0.1774	0.1241	0.0656	0.0142	0.0887
DIR-NB	<u>0.8263</u>	<u>0.8333</u>	-0.0423	0.1659	0.1874	0.0304	0.0215	-1.6500	0.0471
DIR-MLP	0.7360	0.7402	0.0081	0.0649	0.1469	0.1469	0.0820	-0.3818	0.0630
DIR-kNN	0.6570	0.6715	<u>0.0058</u>	0.0468	0.1032	0.1348	0.0564	-0.6333	0.0668
LFR-DT	0.5055	0.5637	-0.0154	0.0384	0.0384	0.0	0.0	NaN	0.0897
LFR-NB	0.5356	0.5882	-0.0184	0.0913	0.1180	0.3333	0.0267	NaN	0.0487
LFR-MLP	0.5523	0.6030	-0.0419	0.1526	0.1793	0.2857	0.0266	NaN	0.0410
LFR-kNN	0.5	0.5588	0.0	0.0	0.0	0.0	0.0	NaN	0.0036
AdaFair	0.8579	0.8529	0.1016	0.0216	0.1068	0.0376	0.0851	0.2250	0.0500
Agarwal's	0.7851	0.7990	0.0366	0.0180	<u>0.0272</u>	0.0504	0.0092	-0.7500	<u>0.0268</u>
EOP-DT	0.7646	0.7696	0.0840	0.0962	0.1167	0.0536	0.0205	0.2190	NaN
EOP-NB	0.7628	0.7794	0.0530	0.0637	0.0729	0.0647	0.0092	-0.5357	NaN
EOP-MLP	0.6938	0.7010	0.1373	0.1334	0.2339	<u>0.0052</u>	0.1005	0.7143	NaN
EOP-kNN	0.6280	0.6421	0.0203	0.1238	0.2048	0.1948	0.0810	-0.4559	NaN
CEP-DT	0.7646	0.7696	0.0840	0.0961	0.1166	0.0536	0.0205	0.2190	NaN
CEP-NB	0.7573	0.7745	0.0458	0.0481	0.0573	0.0616	0.0092	-0.6786	NaN
CEP-MLP	0.6982	0.7059	0.1075	0.1334	0.1816	0.0572	0.0482	0.3947	NaN
CEP-kNN	0.6304	0.6471	0.0972	0.2007	0.2048	0.1731	<u>0.0041</u>	<u>0.1830</u>	NaN

eters provided by the Scikit-learn toolkit¹⁷ since the primary goal of this study is to compare the relative behavior of fairness-aware models under the same experimental setting rather than to optimize predictive performance. As a result, all models are evaluated under the same train-test split to ensure a fair and consistent comparison. Regarding fairness-aware ML models, we use the implementation provided by Iosifidis and Ntoutsis (2019) for the AdaFair method,¹⁸ while the AI Fairness 360 toolkit¹⁹ is used to execute the Agarwal, LFR, DIR, EOP, and CEP methods. In addition, we combine the use of pre-processing and post-processing models with the traditional approaches. In detail, the resulting datasets of pre-processing models, i.e., fair dataset, will be the input of traditional models. Similarly, the outcome of traditional models will be processed by the post-processing fairness-aware models in order to mitigate bias and achieve fairness in the final output. *Sex* is selected as the protected attribute for all datasets due to its popularity.

We report the prediction performance of classification models for each dataset in the Accuracy (Acc.) and Balanced Accuracy (BA) measures because most datasets are imbalanced, as demonstrated in the imbalance ratio (IR) column of Table 2.

¹⁷ <https://scikit-learn.org/>

¹⁸ <https://github.com/iosifidisvasileios/AdaFair>.

¹⁹ <https://github.com/Trusted-AI/AIF360>

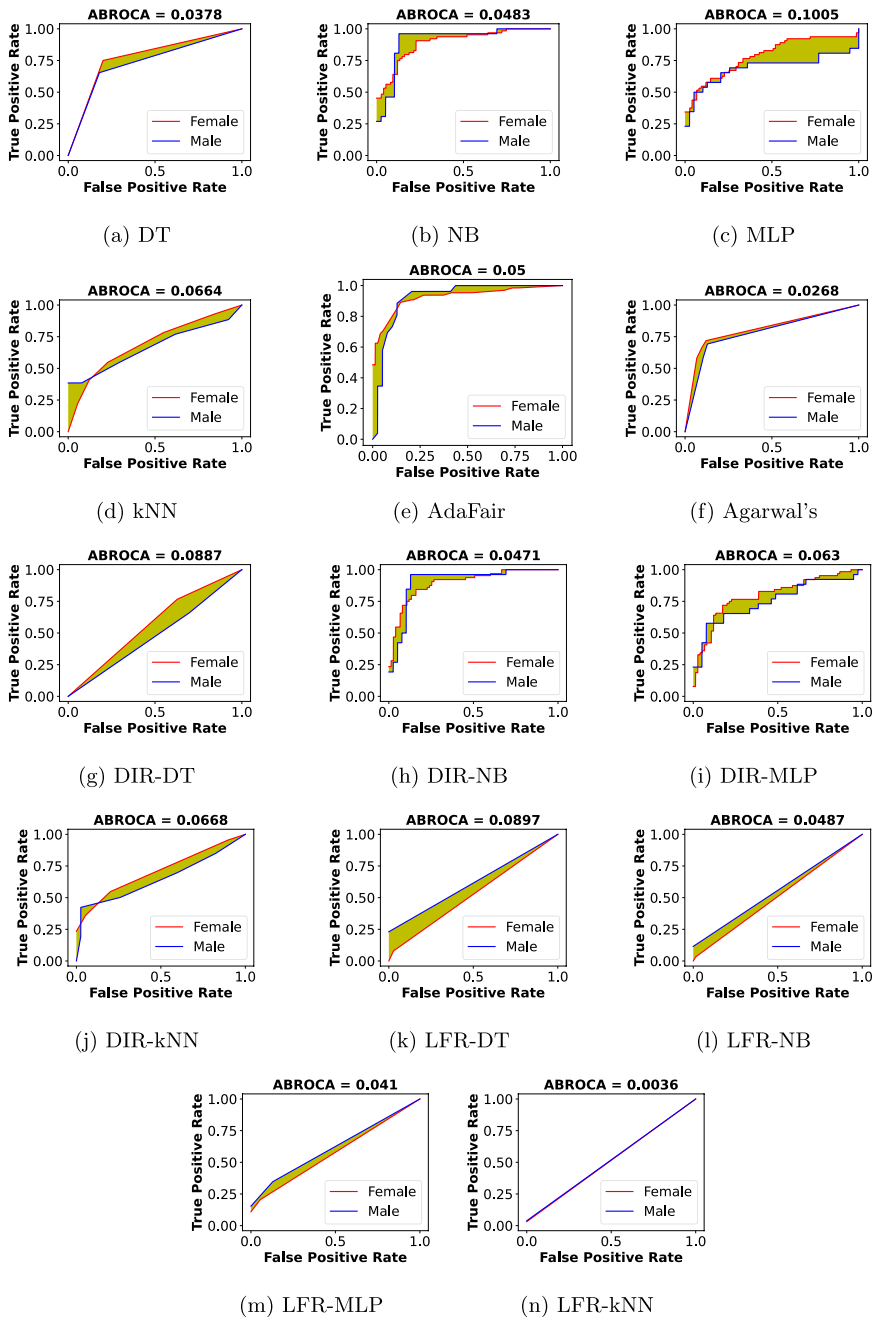


Fig. 7 Credit approval: ABROCA slice plots

Table 4 Credit card clients: performance of predictive models. Protected attribute: Sex

Model	BA	Acc.	SP	EO	EOd	PP	PE	TE	ABROCA
DT	0.6131	0.7277	0.0308	0.0263	0.0656	0.0275	0.0393	0.0071	0.0324
NB	0.5599	0.3778	-0.0034	0.0308	0.0311	0.0226	0.0002	-0.0184	0.0238
MLP	0.6111	0.5782	0.0523	0.0403	0.0883	0.0231	0.0481	0.0148	0.0193
kNN	0.5435	0.7530	0.0153	0.0089	0.0230	0.0105	0.0142	0.0454	0.0115
DIR-DT	0.6099	0.7187	0.0290	<u>0.0034</u>	0.0342	0.0011	0.0308	-0.0079	0.0174
DIR-NB	0.5674	0.4104	0.0121	0.0174	0.0333	0.0215	0.0158	-0.0152	0.0240
DIR-MLP	0.5301	<u>0.7814</u>	0.0245	0.0258	0.0479	0.1022	0.0220	7.9436	0.0129
DIR-kNN	0.5471	0.7511	0.0053	0.0106	<u>0.0111</u>	0.0482	<u>0.0005</u>	-0.3479	0.0101
LFR-DT	0.5798	0.5897	0.0476	0.0039	0.0586	0.0061	0.0546	<u>-0.0043</u>	0.0062
LFR-NB	0.4831	0.7103	-0.0053	0.0081	0.0161	0.0457	0.0079	-0.5013	<u>0.0063</u>
LFR-MLP	0.4514	0.6406	0.0117	0.0264	0.0373	0.0384	0.0109	-0.8892	0.0074
LFR-kNN	0.4967	0.2270	-0.0044	0.0051	0.0091	0.0267	0.0040	-0.0026	0.0106
AdaFair	0.6460	0.8160	0.0062	0.0321	0.0391	0.0155	0.0070	-0.2601	0.0094
Agarwal's	0.5025	0.5270	0.0045	0.0228	0.0238	0.0362	0.0009	-0.0392	0.0098
EOP-DT	<u>0.6132</u>	0.7278	0.0305	0.0263	0.0652	0.0271	0.0389	0.0056	NaN
EOP-NB	0.5548	0.3714	0.0090	0.0379	0.0566	0.0165	0.0187	-0.0199	NaN
EOP-MLP	0.6073	0.5812	0.0138	0.0026	0.0137	0.0267	0.0110	-0.0311	NaN
EOP-kNN	0.5416	0.7534	0.0062	0.0053	0.0119	0.0107	0.0066	-0.2388	NaN
CEP-DT	0.6131	0.7277	0.0308	0.0263	0.0656	0.0275	0.0393	0.0071	NaN
CEP-NB	0.5599	0.3778	<u>-0.0035</u>	0.0308	0.0311	0.0226	0.0002	-0.0184	NaN
CEP-MLP	0.6111	0.5782	0.0522	0.0403	0.0883	0.0231	0.0481	0.0148	NaN
CEP-kNN	0.5407	0.7561	0.0302	0.0322	0.0590	<u>0.0051</u>	0.0268	0.6475	NaN

5.2 Experimental results

5.2.1 Credit approval dataset

The experimental results on the Credit approval dataset are presented in Table 3 and Fig. 7. The dashed lines are used to separate the group of predictive models (traditional, pre-processing, in-processing, and post-processing approaches). The best value is shown in **bold**, and the second-best value is underlined.

The results indicate that classification models cannot satisfy multiple fairness measures simultaneously. Among fairness-aware models, AdaFair is the best model in terms of accuracy (0.8529) and balanced accuracy (0.8579). Besides, LFR-kNN is a notable model with the best performance on SP, EO, EOd, PE and ABROCA measures. However, its accuracy and balanced accuracy are very low with values 0.5588 and 0.5, respectively. In addition, most pre-processing models prioritize fairness at the cost of significant reductions in accuracy and balanced accuracy. For the TE measures, as defined in Eq. 6, the result may be “NaN” if the denominator equals zero. In addition, a limitation of the post-processing models is the inability to obtain ABROCA values, as calculating ABROCA requires model probabilities across multiple thresholds. This is reflected in “NaN” values in Table 3. Overall, the in-processing approach outperforms pre-processing and post-processing approaches in terms of trade-off between predictive performance and fairness constraints.

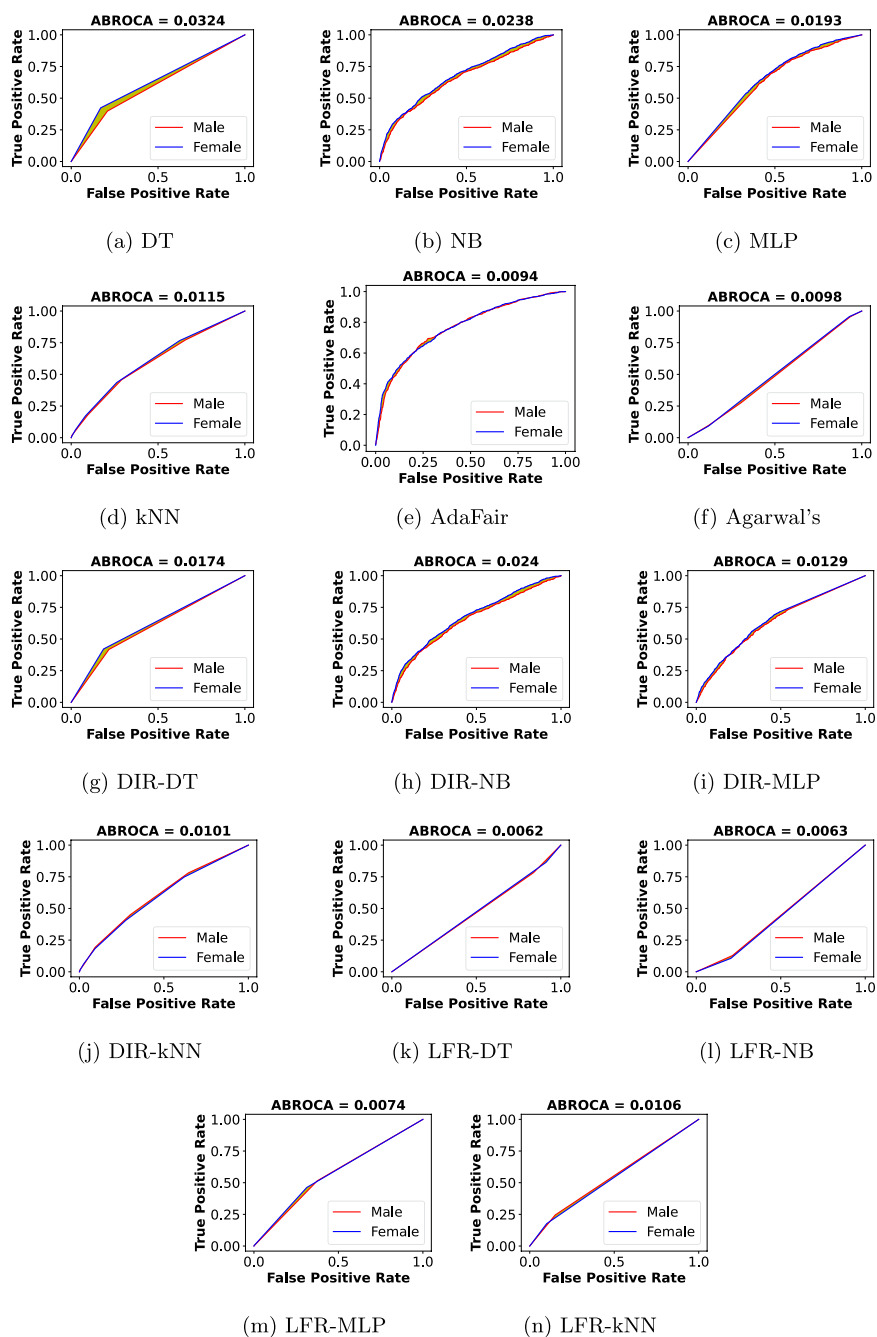


Fig. 8 Credit card: ABROCA slice plots

Table 5 Credit scoring: performance of predictive models. Protected attribute: Sex

Model	BA	Acc.	SP	EO	EOd	PP	PE	TE	ABROCA
DT	0.9761	0.9924	0.0345	0.0032	0.0269	0.0003	0.0238	0.8333	0.0132
NB	0.8785	0.9585	0.0448	0.0149	0.1126	0.0003	0.0976	0.7148	0.0088
MLP	<u>0.9923</u>	0.9931	0.0299	<u>0.0001</u>	0.0124	0.0013	0.0122	NaN	0.0005
kNN	0.8447	0.9581	0.0460	0.0236	0.0596	0.0082	0.0359	0.7267	0.0238
DIR-DT	0.9713	0.9908	0.0361	0.0009	0.0764	0.0043	0.0754	1.0286	0.0377
DIR-NB	0.9612	0.9292	0.0934	0.0702	0.0702	0.0	0.0	NaN	0.0064
DIR-MLP	0.9820	0.9924	0.0284	<u>0.0001</u>	0.0218	0.0029	0.0217	−3.6667	<u>0.0006</u>
DIR-kNN	0.8378	0.9562	0.0503	0.0267	0.0868	0.0068	0.0601	0.8279	0.0222
LFR-DT	0.4674	0.7632	<u>0.0143</u>	0.0191	0.0360	0.0365	0.0169	−0.7024	0.0057
LFR-NB	0.4683	0.8549	−0.0611	0.0642	0.0642	0.0276	0.0	−1.0684	0.0235
LFR-MLP	0.5	0.9128	0.0	0.0	0.0	0.0312	0.0	0.0	0.0644
LFR-kNN	0.4674	0.7632	<u>0.0143</u>	0.0191	0.0360	0.0365	0.0169	−0.7024	0.0232
AdaFair	0.9929	0.9943	0.0298	<u>0.0001</u>	0.0124	0.0014	0.0123	NaN	0.0009
Agarwal's	0.9077	0.9649	0.0421	0.0157	0.0442	0.0038	0.0284	0.7200	0.0405
EOP-DT	0.9761	0.9924	0.0345	0.0032	0.0269	0.0030	0.0238	0.8333	NaN
EOP-NB	0.8598	0.9532	0.0273	0.0012	0.0498	0.0053	0.0486	−0.2118	NaN
EOP-MLP	0.9901	<u>0.9928</u>	0.0293	<u>0.0001</u>	0.0185	0.0020	0.0184	NaN	NaN
EOP-kNN	0.8317	0.9524	0.0311	0.0126	0.0286	0.0129	0.0160	<u>0.1192</u>	NaN
CEP-DT	0.9761	0.9924	0.0345	0.0032	0.0270	0.0034	0.0238	0.8333	NaN
CEP-NB	0.8617	0.9566	0.0561	0.0182	0.2371	0.0081	0.2189	1.0442	NaN
CEP-MLP	0.9901	<u>0.9928</u>	0.0309	<u>0.0001</u>	<u>0.0030</u>	<u>0.0002</u>	<u>0.0029</u>	−1.000	NaN
CEP-kNN	0.8229	0.9543	0.0563	0.0236	0.2111	0.0022	0.1875	0.7728	NaN

5.2.2 Credit card clients dataset

Table 4 and Fig. 8 report the experimental results of the predictive models on the credit card clients dataset. AdaFair once again demonstrates its ability to achieve accurate predictions while maintaining fairness, standing out as a fair classification model with the highest accuracy and balanced accuracy, at 0.8160 and 0.6460, respectively. The pre-processing approaches exhibit strong performance on specific fairness measures: LFR-kNN achieves the best results on EOd and TE, DIR-DT performs best on PP, and LFR-DT attains the lowest ABROCA value. Nevertheless, these models generally yield lower accuracy and balanced accuracy compared to the traditional classification models.

5.2.3 Credit scoring dataset

The experimental results on the credit scoring dataset are presented in Table 5 and Fig. 9. In terms of predictive performance, AdaFair once again outperforms the other models, achieving very high accuracy (0.9943) and balanced accuracy (0.9929). In addition, the pre-processing model LFR-MLP attains perfect scores on multiple fairness measures, including SP, EO, EOd, PE, and TE. However, its balanced accuracy remains low at 0.5. The traditional MLP model stands out by achieving the best performance on the ABROCA fairness measure (0.0005). By contrast, the pre-processing LFR model performs poorly due to its very low balanced accuracy (below 0.5),

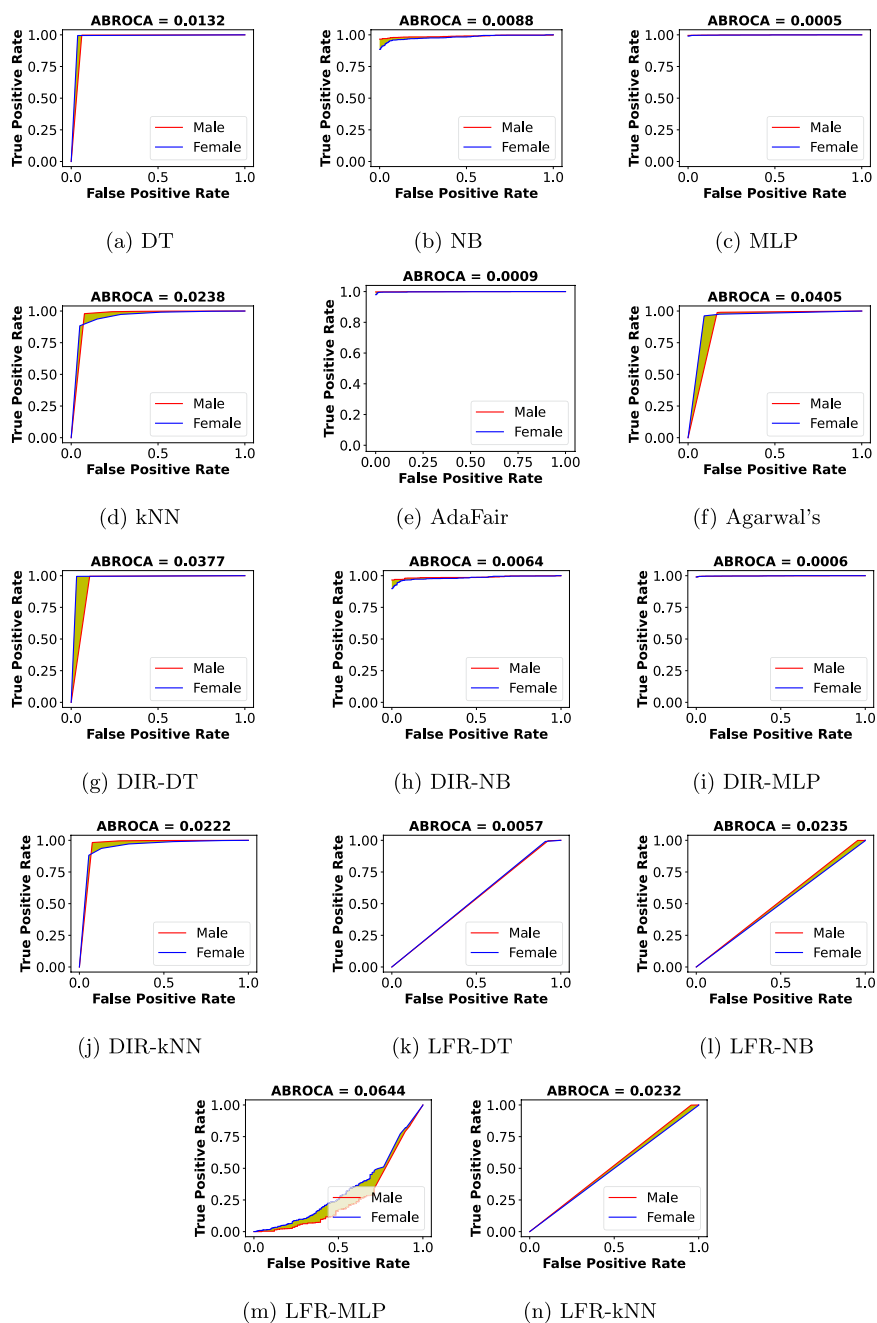


Fig. 9 Credit scoring: ABROCA slice plots

Table 6 German credit: performance of predictive models. Protected attribute: Sex

Model	BA	Acc.	SP	EO	EOd	PP	PE	TE	ABROCA
DT	0.5954	0.6567	0.0485	0.0160	0.1807	0.0292	0.1646	0.0769	0.0903
NB	0.6604	0.7300	<u>0.0019</u>	0.0614	0.1615	0.0166	0.1001	-0.2557	0.1012
MLP	0.6095	0.6634	-0.0669	0.0936	0.1292	0.0214	0.0356	-0.6250	0.0697
kNN	0.5348	0.6500	0.0641	0.0670	0.1171	0.0391	0.0501	0.1399	0.0458
DIR-DT	0.6221	0.6767	-0.0736	0.0972	0.1489	0.0263	0.0517	-0.6653	<u>0.0227</u>
DIR-NB	<u>0.6392</u>	0.7133	-0.0094	0.0511	0.0983	<u>0.0043</u>	0.0473	-0.2667	0.0970
DIR-MLP	0.5676	0.7000	-0.0326	0.0625	<u>0.0781</u>	0.0169	<u>0.0156</u>	-0.2178	0.1114
DIR-kNN	0.5118	0.6267	-0.0144	0.0608	0.1431	0.0018	0.0823	-0.2114	0.1223
LFR-DT	0.5686	0.5933	-0.0174	0.0646	0.1325	0.0128	0.0679	-0.3510	0.0032
LFR-NB	0.4861	0.5433	-0.0592	0.0410	0.1361	0.0546	0.0950	-0.5092	0.0342
LFR-MLP	0.5	0.6967	0.0	0.0	0.0	0.0371	0.0	0.0	0.0545
LFR-kNN	0.5467	0.6667	-0.0570	0.0649	0.1161	0.0336	0.0512	-0.2826	0.0434
AdaFair	0.5554	0.7133	-0.0442	0.0364	0.1104	0.0453	0.0740	-0.1184	0.0801
Agarwal's	0.6289	0.7033	-0.0945	0.1116	0.2001	0.0350	0.0884	-0.6364	0.0384
EOP-DT	0.5954	0.6567	0.0485	0.0160	0.1807	0.0292	0.1646	0.0769	NaN
EOP-NB	0.6286	0.6900	-0.0935	0.1347	0.1703	0.0131	0.0356	-0.7500	NaN
EOP-MLP	0.5990	0.6533	-0.0877	0.1069	0.1770	0.0297	0.0701	-0.7279	NaN
EOP-kNN	0.5309	0.6533	0.0877	0.0870	0.1693	0.0357	0.0823	0.2190	NaN
CEP-DT	0.5954	0.6567	0.0485	0.0160	0.1807	0.0292	0.1646	0.0769	NaN
CEP-NB	0.6153	<u>0.7233</u>	0.1151	<u>0.0120</u>	0.3218	0.0597	0.3098	0.1980	NaN
CEP-MLP	0.5667	0.6600	0.0510	0.0136	0.1877	0.0227	0.1741	<u>0.0208</u>	NaN
CEP-kNN	0.5257	0.6633	0.1396	0.1337	0.2805	0.0307	0.1468	0.3783	NaN

whereas the DIR model demonstrates a more favorable balance by achieving both accurate classification performance and strong fairness outcomes.

5.2.4 German Credit Dataset

Table 6 and Fig. 10 present the experimental results of the predictive models on the German Credit dataset. In terms of classification performance, the traditional NB model achieves the best results, with an accuracy of 0.7300 and a balanced accuracy of 0.6604. With respect to fairness constraints, LFR-MLP emerges as the top-performing model, achieving perfect scores on multiple fairness measures, including SP, EO, EOd, PE, and TE. However, its balanced accuracy remains low at 0.5. Among the pre-processing approaches, DIR proves effective in achieving a more favorable trade-off between predictive performance and fairness. Although the post-processing models produce relatively fair classification outcomes, they do not demonstrate strong performance with respect to the fairness constraints.

5.2.5 PAKDD Credit Dataset

In the PAKDD credit risk assessment dataset, all methods exhibit low balanced accuracy, as shown in Table 7 and Fig. 11. Among the in-processing approaches, AdaFair achieves superior performance, obtaining the best results on 6 out of 9 evaluation measures, including accuracy and five fairness measures. In the pre-processing

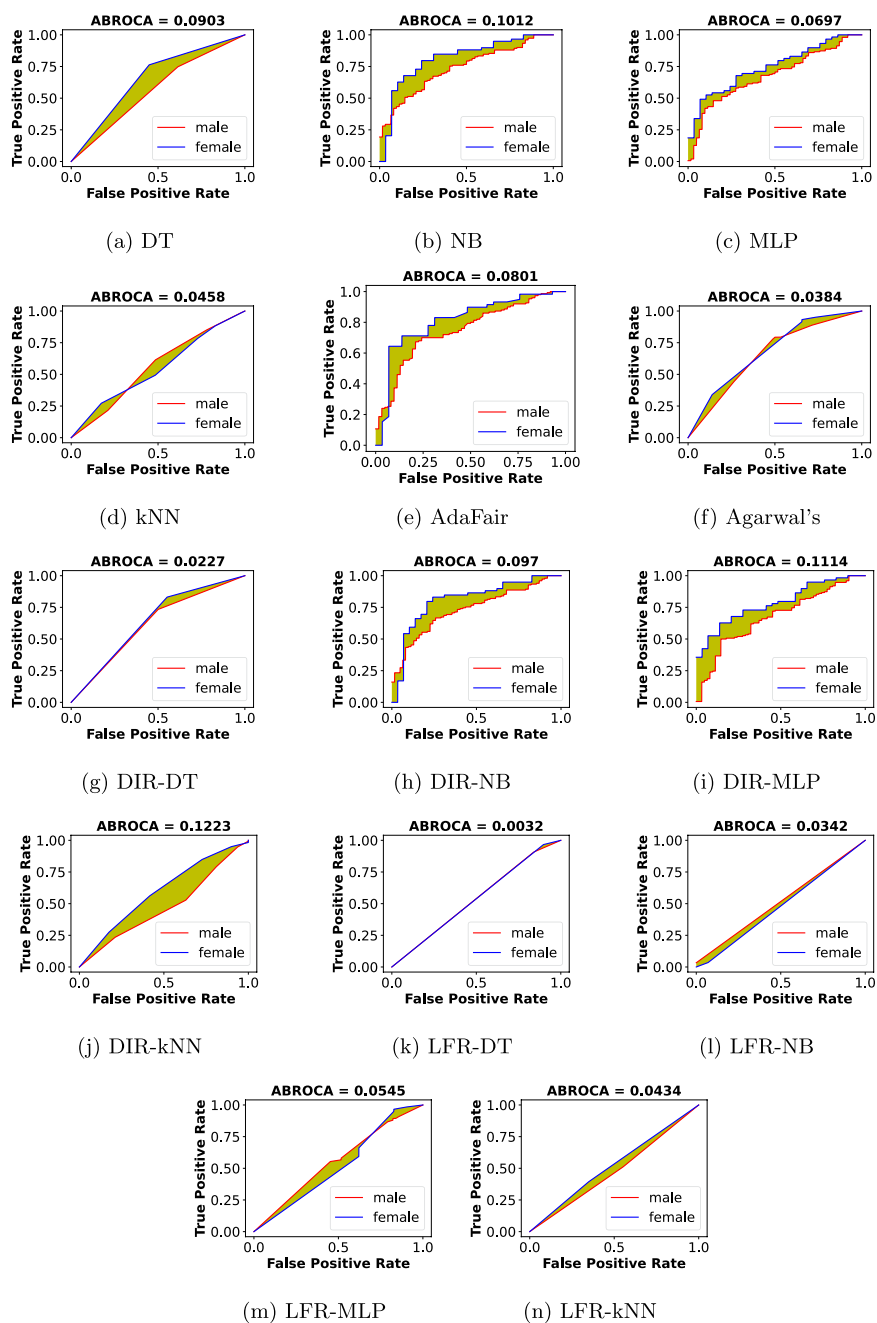


Fig. 10 German credit: ABROCA slice plots

Table 7 PAKDD credit: performance of predictive models. Protected attribute: Sex

Model	BA	Acc.	SP	EO	EOd	PP	PE	TE	ABROCA
DT	0.5241	0.6244	0.0124	0.0325	0.0358	0.0476	0.0033	-0.0707	0.0146
NB	0.5088	0.7256	0.0022	0.0087	0.0143	0.0523	0.0056	0.3997	0.0110
MLP	0.5119	0.6925	0.0655	0.0715	0.1340	0.0224	0.0624	1.5546	0.0144
kNN	0.5057	0.6822	-0.0056	0.0192	0.0201	<u>0.0013</u>	0.0010	-0.4404	0.0094
DIR-DT	<u>0.5174</u>	0.6189	0.0253	0.0116	0.0493	0.0090	0.0377	<u>-0.0109</u>	0.0247
DIR-NB	0.5130	0.7210	0.0251	0.0155	0.0432	0.0556	0.0277	3.4949	0.0112
DIR-MLP	0.5003	<u>0.7351</u>	0.0	0.0018	<u>0.0024</u>	0.5833	<u>0.0006</u>	-909.4	0.0128
DIR-kNN	0.5027	0.6810	-0.0028	0.0044	0.0101	0.0437	0.0056	-0.4816	0.0134
LFR-DT	0.4949	0.7200	0.0069	<u>0.0013</u>	0.0106	0.0189	0.0093	2.6914	<u>0.0040</u>
LFR-NB	0.5034	0.2771	-0.0088	0.0064	0.0163	0.0271	0.0100	-0.0033	0.0017
LFR-MLP	0.4986	0.7180	0.0034	0.0070	0.0145	0.0543	0.0074	0.6436	0.0072
LFR-kNN	0.5075	0.4736	0.0212	0.0024	0.0319	0.0155	0.0294	-0.0239	0.0159
AdaFair	0.5	0.7353	0.0	0.0	0.0	0.0	0.0	NaN	0.0145
Agarwal's	0.5093	0.7263	-0.0017	0.0140	0.0162	0.0549	0.0021	-0.9031	0.0081
EOP-DT	0.5241	0.6244	0.0124	0.0325	0.0358	0.0476	0.0033	-0.0707	NaN
EOP-NB	0.5083	0.7258	<u>-0.0009</u>	0.0134	0.0165	0.0613	0.0031	-0.5122	NaN
EOP-MLP	0.5132	0.6854	0.0120	0.0123	0.0233	0.0270	0.0110	-0.0755	NaN
EOP-kNN	0.5057	0.6817	-0.0076	0.0214	0.0242	<u>0.0013</u>	0.0029	-0.4848	NaN
CEP-DT	0.5241	0.6244	0.0124	0.0325	0.0358	0.0476	0.0033	-0.0707	NaN
CEP-NB	0.5088	0.7269	0.0068	0.0037	0.0138	0.0696	0.0101	2.6232	NaN
CEP-MLP	0.5127	0.7029	0.0996	0.1051	0.2018	0.0069	0.0967	4.6288	NaN
CEP-kNN	0.5060	0.6860	0.0069	0.0070	0.0186	0.0053	0.0116	-0.1113	NaN

group, although LFR-NB achieves improved fairness in classification outcomes (as measured by TE and ABROCA), this comes at the cost of a significant reduction in accuracy. In addition, the post-processing models outperform the other approaches in terms of balanced accuracy.

5.3 Discussion and limitations

Across all datasets, we observe that different fairness-aware approaches exhibit distinct trade-offs between predictive performance and fairness constraints. As expected, fairness-aware models generally achieve better results with respect to fairness measures compared to traditional models. Among these, LFR-MLP and AdaFair are notable methods that outperform others across multiple datasets with respect to fairness constraints. However, LFR-MLP sacrifices predictive accuracy to improve fairness. Interestingly, AdaFair demonstrates outstanding capability by improving not only accuracy but also fairness in the results across three datasets. Furthermore, the results demonstrate that different fairness measures may lead to different conclusions about the fairness of the model. This finding highlights the importance of evaluating fairness-aware models under multiple fairness measures in financial decision-making systems.

The limitations, which can be improved in further research, are following. Firstly, the evaluation of fairness in this study is limited to individual protected attributes and does not yet consider the simultaneous impact of multiple protected attributes, such

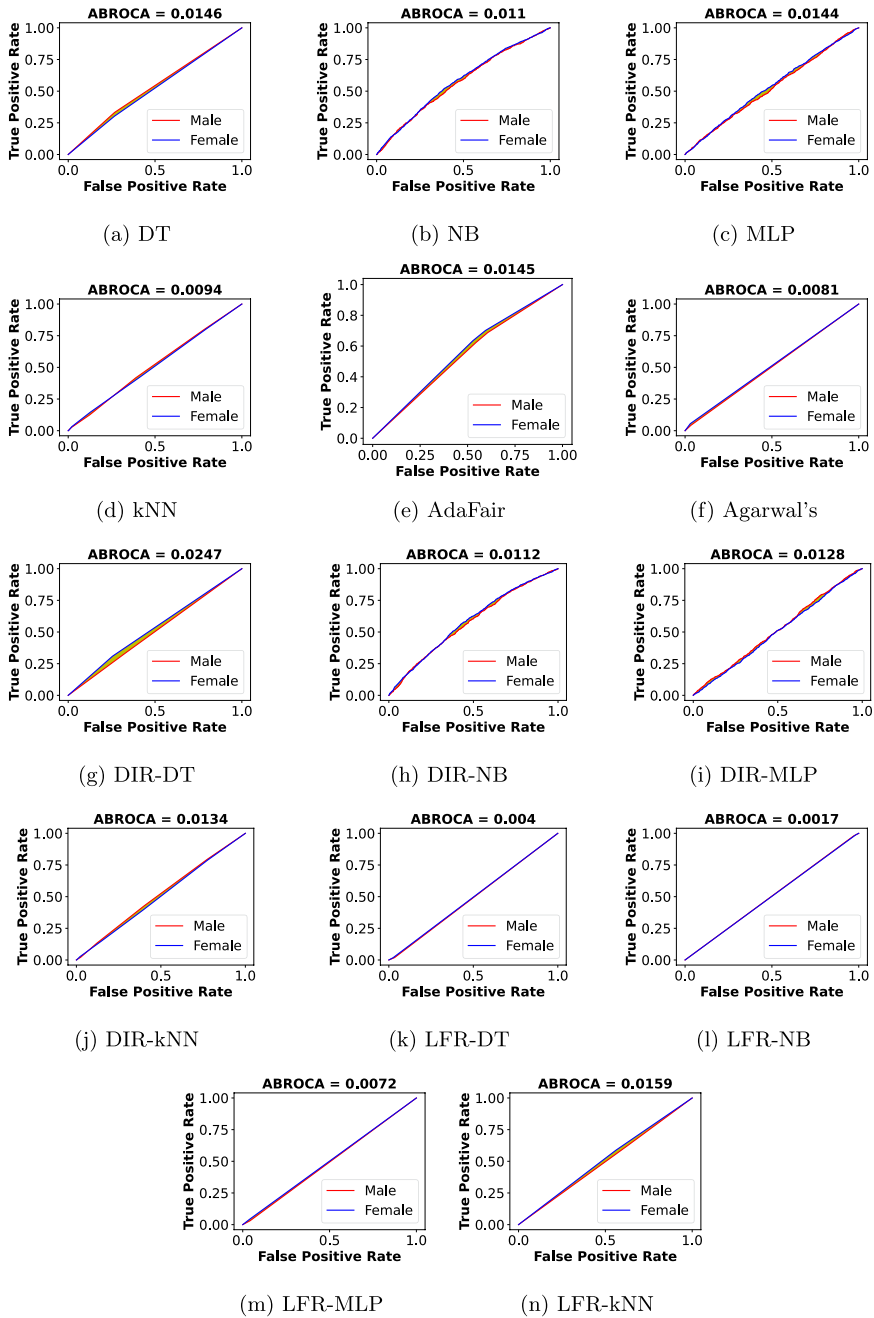


Fig. 11 PAKDD credit: ABROCA slice plots

as gender and race, nor the relationships among different fairness measures as well as the intersection fairness. Secondly, our analysis relies on commonly used real-world datasets, which may not fully capture complex or domain-specific biases, particularly in financial applications where high-quality and unbiased datasets are difficult to obtain. Another limitation of this study is that the experiments are conducted using a single train-test split without repeated runs or cross-validation. Finally, while this study focuses on fairness assessment, it does not explicitly address the development of fair or explainable classification models, limiting insights into the underlying sources of bias within both the learning algorithms and the datasets.

6 Conclusions and outlook

In this work, we investigated commonly used credit scoring datasets in ML for financial applications. Using the BN, we analyzed structural dependencies between protected attributes and the class label, providing insights into statistical relationships present in the datasets. Afterward, we evaluated traditional classifiers in comparison with various fairness-aware models across three approaches: pre-processing, in-processing, and post-processing. The experimental results show that fairness-aware methods can improve fairness measures while maintaining competitive predictive performance compared to traditional models. These findings provide practical insights for financial institutions when selecting fairness-aware models for automated credit scoring systems. Moreover, these results highlight the importance of examining both dataset characteristics and model behavior when studying fairness in credit scoring.

In the future, we plan to expand the fairness evaluation to simultaneously address multiple protected attributes, such as gender and race, while further exploring the correlations between different fairness measures. Additionally, understanding commonly used datasets motivates us to research and develop fair synthetic data generation models for the finance domain, as finding a perfect dataset in the real world has never been a straightforward task. Furthermore, we aim to develop fair and explainable classification algorithms to understand the root causes of bias within the algorithms themselves as well as in the datasets.

Acknowledgements We sincerely thank the anonymous reviewers and editors for their valuable comments and suggestions, which have greatly improved the quality and clarity of this paper.

Author contributions All authors contributed to the conception and design of the study. Material preparation, data collection and analysis were performed by Huyen Giang Thi Thu, Thang Viet Doan and Tai Le Quy. The first draft of the manuscript was written by Huyen Giang Thi Thu, Ha-Bang Ban and Tai Le Quy and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding Open Access funding enabled and organized by Projekt DEAL. The authors did not receive support from any organization for the submitted work.

Data availability The source code and dataset are publicly available at <https://github.com/tailequy/faircredit>

Code availability The source code and dataset are publicly available at <https://github.com/tailequy/faireredit>.

Declarations

Conflict of interest The authors declare no conflict of interest.

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